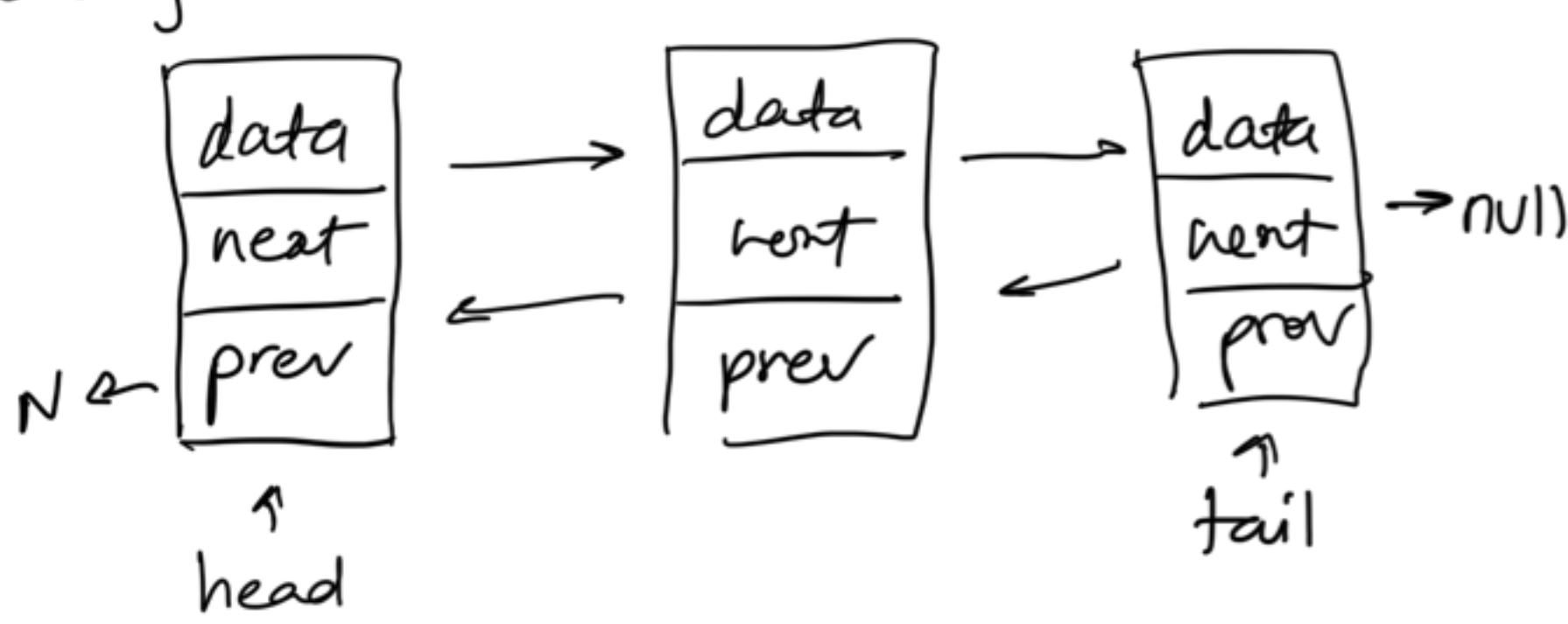


Doubly linked list



```

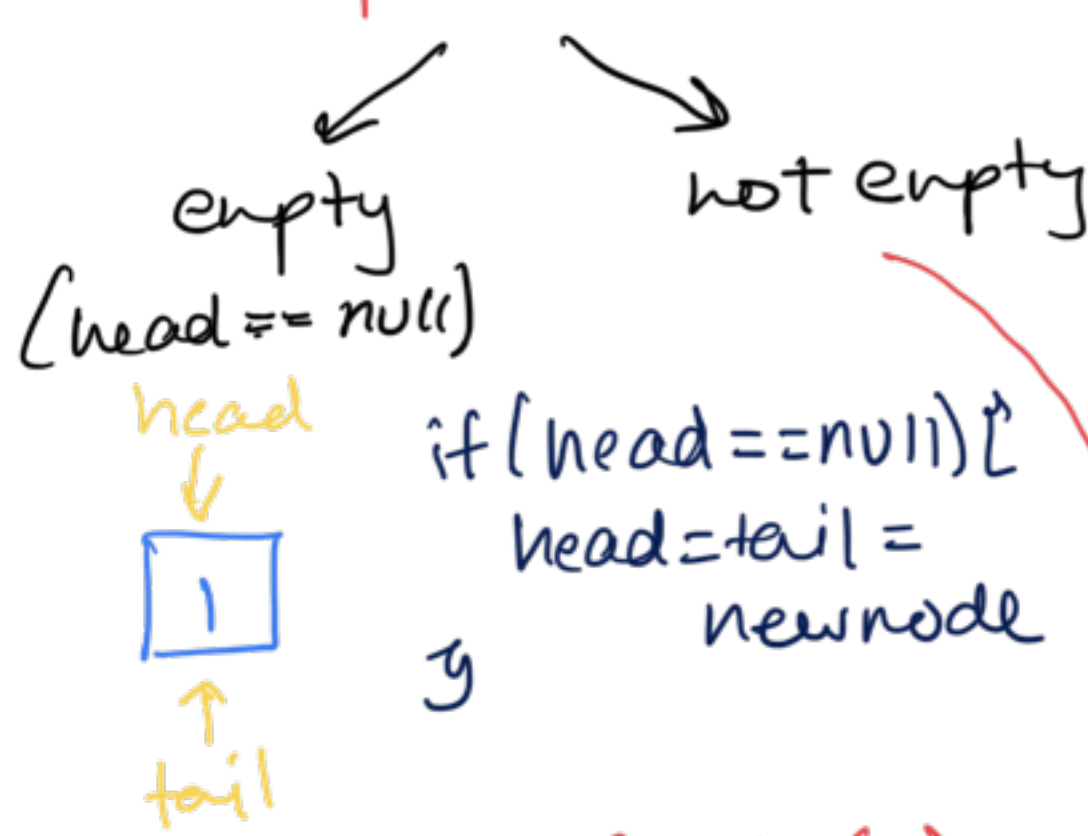
class Node {
    int data;
    Node * next;
    Node * prev;
}

class Doublylist {
    Node * head;
    Node * tail;
}

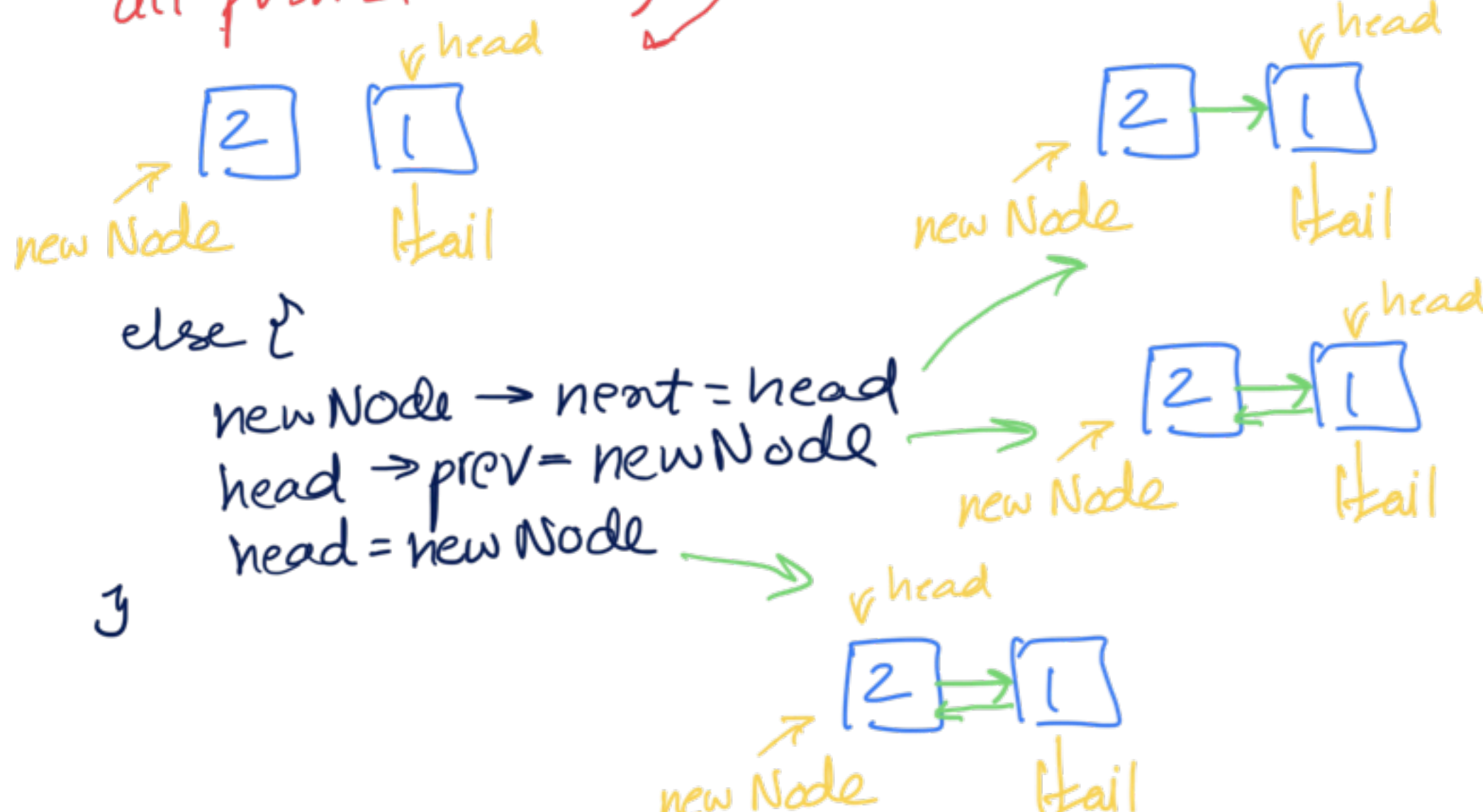
Node * newNode = new Node(val);
    
```

① Push front

dll.push-front(1)



dll.push-front(2)



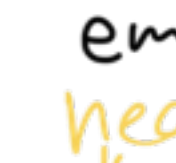
```

if (head == null) {
    head = tail =
    newnode;
}
else {
    newNode -> next = head;
    head -> prev = newNode;
    head = newNode;
}
    
```

② Push back

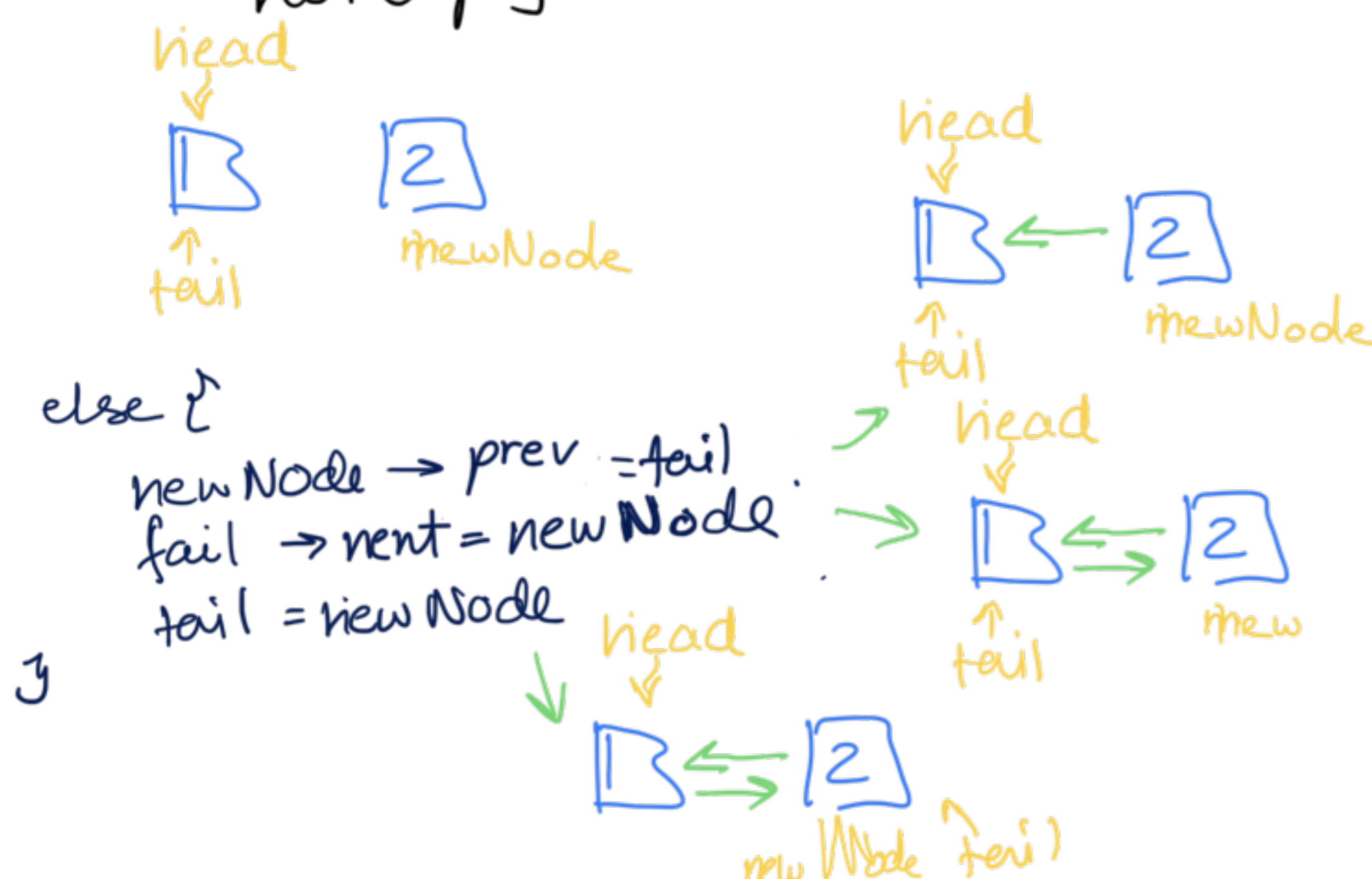
dll.push-back(1)

empty



dll.push-back(2)

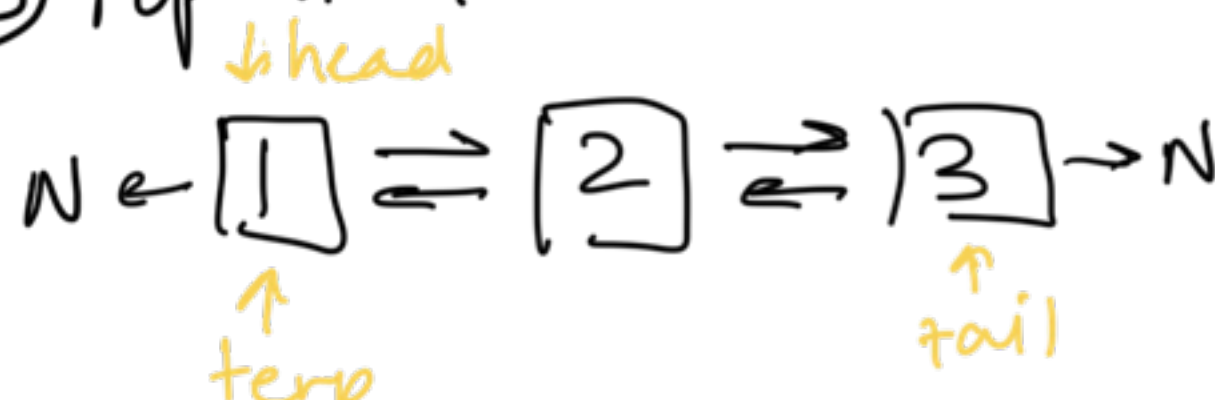
not empty



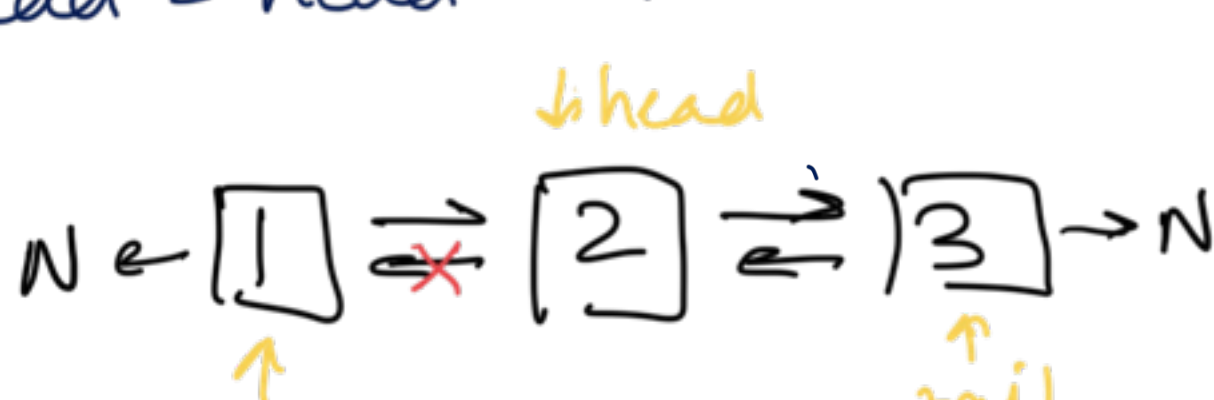
```

else {
    newNode -> prev = tail;
    tail -> next = newNode;
    tail = newNode;
}
    
```

③ Pop front



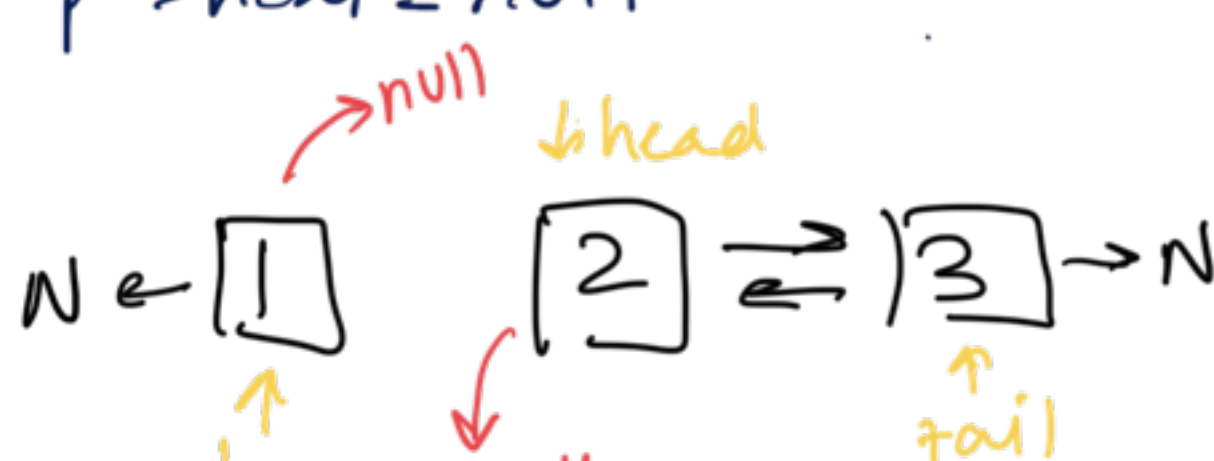
Node *temp = head
head = head -> next



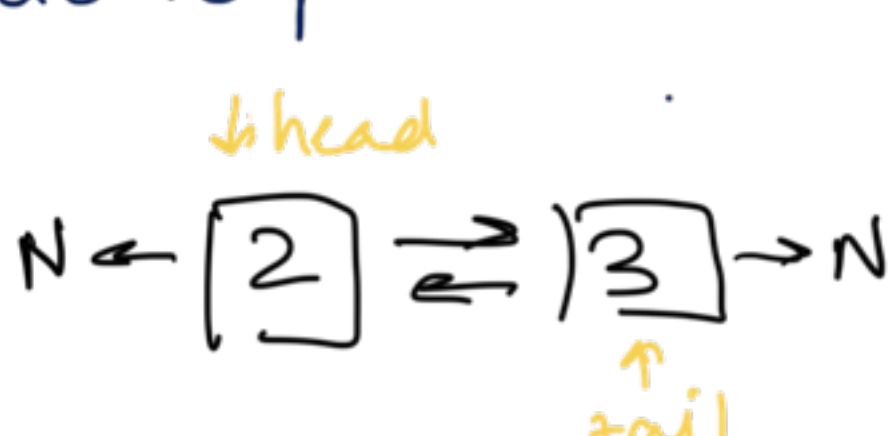
if (head != null)
head -> prev = null



temp -> next = null



delete temp



```

Node *temp = head;
head = head -> next;
if (head != null)
    head -> prev = null;
temp -> next = null;
delete temp;
    
```

④ Pop back

```

Node *temp = tail;
tail = tail -> prev;
if (tail != null)
    tail -> next = null;
temp -> prev = null;
delete temp;
    
```